

Chapter 02

Data profiling

Agenda



- Concepts
- The Profiling Process
- Data Quality Dimensions
- Tools & Techniques
- Missing Data

"Garbage In, Garbage Out"

- **The Golden Rule:**

"Garbage In, Garbage Out" (GIGO) - Concept attributed to George Fuechsel (IBM), late 1950s.

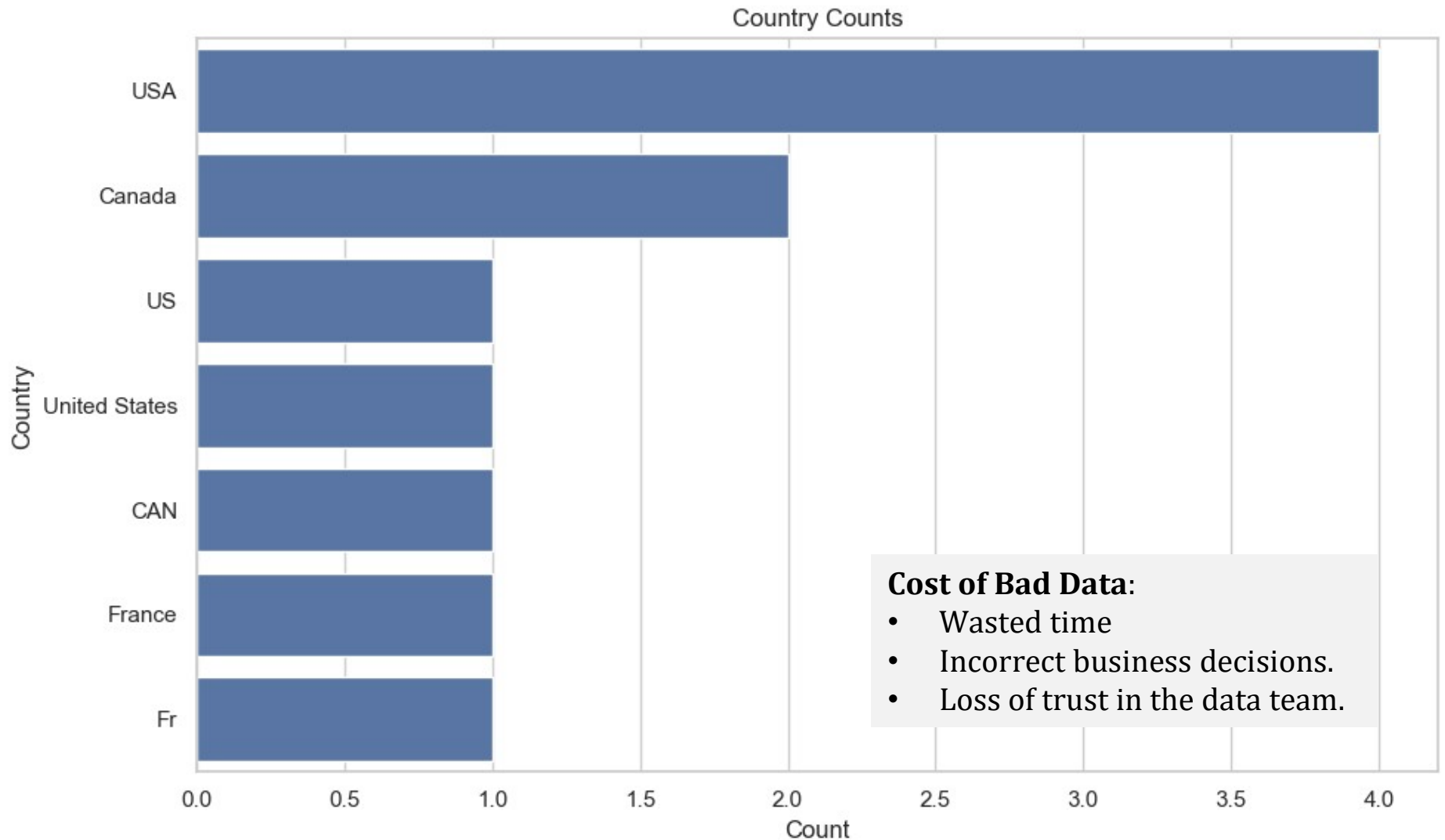
"Garbage In, Garbage Out"

ID	Name	Age	Email	RegistrationDate	Salary	Gender	Country
1	Alice	25	alice@example.com	2023-01-01	50000	F	USA
2	Bob	30	bob@invalid	01/02/2023	60000	M	Canada
3	Charlie	-5		2023-13-01	-1000	Male	US
4	David	40	david@example.com	2023-01-01	70000	F	Canada
1	Alice	25	alice@example.com	2023-01-01	50000	F	USA
6	Eve	150	eve@		8000000	Female	
7	Frank		frank@example.com	Feb 3, 2023			United States
8	Grace	35		2023-01-01	90000	Unknown	CAN
9	Hank	twenty	hank@domain.	2023/01/01	fifty k	M	USA
10	Ivy	28	ivy@example.com	2023-01-01	100000	F	France
11	Jack	0	jack@invalid	invalid date	0	Other	Fr
12	Alice	500	alice@	2023-01-32		M	USA

"Garbage In, Garbage Out"

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"Garbage In, Garbage Out"



Definition **data profiling**



- **Data profiling** - refers to the process of examining, analyzing, reviewing and summarizing data sets to gain insight into the quality of data
- **Types of data profiling**
 - ***Structure discovery.*** This focuses on the formatting of the data, making sure everything is uniform and consistent. It uses basic statistical analysis to return information about the validity of the data.
 - ***Content discovery.*** This process assesses the quality of individual pieces of data. For example, ambiguous, incomplete and null values are identified.
 - ***Relationship discovery.*** This detects connections, similarities, differences and associations among data sources.

Benefits of data profiling

- ***Understand data quality***

- accuracy, consistency, completeness, and reliability of the data.

- ***Identify data issues***

- identify issues like missing or inconsistent data, duplicate entries, invalid data, or data that does not conform to your business rules or standards

- ***Understand data structure and relationships***

- provide insights into how data is structured and how different data elements relate to each other

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The Profiling Process



- **Structure Discovery**
 - Is the data formatted correctly?
- **Content Discovery**
 - Does the data make sense?
- **Relationship Discovery**
 - How does the data connect?

Structure Discovery

- *Focus*: Formatting and consistency.
- *Checks*
 - *Pattern Matching*: Are phone numbers consistent?
 - *Data Types*: Is 'Age' stored as a Number or Text?
 - *Format Constraints*: Date formats
- **Example:**
 - *Bad*: '2023-01-01', '01/02/2023', 'Jan 1st 23'
 - *Good*: '2023-01-01', '2023-02-01', '2023-01-01'

Content Discovery

- *Focus*: The actual values inside the cells.
- *Key Statistics*:
 - *Null Count*: How many missing values?
 - *Distinct Count* (Cardinality): How many unique values?
 - *Min / Max / Average*: For numerical fields.
 - *Mode*: Most frequent value.
- **Example**
 - Column 'Age':
 - Min: -5 (Error!)
 - Max: 200 (Error!)
 - Mean: 24.5 (Reasonable)

Relationship Discovery

- *Focus*: Links between data points
- *Check*
 - *Key Dependencies*: Primary Keys
 - *Referential Integrity*: Foreign Keys.
 - *Functional Dependencies*: Logical rules
 - Example: If 'City' is "Paris", 'Country' MUST be "France".

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Data Quality Dimensions



- **6 Core Dimensions** (*DAMA-DMBOK*)
 - Completeness
 - Uniqueness
 - Timeliness
 - Validity
 - Accuracy
 - Consistency

Completeness & Uniqueness



- **Completeness**

- *Definition:* Is all required data present?
- *Measure:* % of Null/Blank values.
- Example: "The 'Phone Number' field is only 60% complete."

- **Uniqueness:**

- *Definition:* Are there duplicates?
- *Measure:* % of duplicate rows or IDs
- Example: "Customer 'John Doe' appears twice with the same ID."

Timeliness & Validity



- **Timeliness**

- *Definition:* Is the data available when needed / Is it up to date
- *Measure:* Time gap between event and data availability.
- Example: "Stock market data arrived 15 minutes delayed."

- **Validity:**

- *Definition:* Does the data follow the defined rules (format, type, range)?
- *Measure:* % of failing records against rules.
- Example: "Email 'john.doe' is invalid (missing @domain.com)."

Accuracy & Consistency



- **Accuracy**
 - *Definition:* Does the data reflect the real world?
 - *Difference from Validity:* A valid phone number (correct format) can be Inaccurate (it's the wrong person's number).
 - *Verification:* Often requires external reference ("Gold Standard").
- **Consistency:**
 - *Definition:* Is the data the same across different systems?
 - Example: CRM says Customer is "Active", but Billing System says "Churned".

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Data Profiling Tools

• Python / Developer Tools

- *YData Profiling*: Automates EDA with one line of code. Generates a complete report (HTML) with missing values, correlations, and distributions.
 - Source: <https://github.com/ydataai/ydata-profiling>
- *Great Expectations*: Best for distinct data quality testing and validation rules.
 - Source: <https://greatexpectations.io/>

• No-Code / Online Tools (Free):

Tool Name	Introduction	Link
Rows	Rows offers built-in data profiling with automatic summaries of column types, missing values, and distributions in a spreadsheet-like interface.	https://rows.com/
Grist	Grist enables basic data profiling through interactive tables, filters, and formulas to explore data structure and consistency.	https://www.getgrist.com/
StatSim Profile	StatSim provides fast, in-browser CSV profiling with instant statistics on missing values, frequencies, and distributions.	https://statsim.com/profile
Excel	Microsoft Excel provides basic data profiling using formulas, filters, and PivotTables to summarize missing values and distributions.	

Common Data Defects

ID	Name	Age	Email	RegistrationDate	Salary	Gender	Country
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Common Data Defects

- Missing Values (Completeness):
 - Blank cells (NaN/null) in 'Email', 'Age', 'RegistrationDate'.
- Duplicates (Uniqueness):
 - Repeated IDs (ID 1).
- Outliers (Validity/Accuracy):
 - Extreme values like 'Age=150' or 'Age=500', 'Salary=8,000,000'.
- Invalid Values (Validity):
 - Age=-5 (Negative)
 - RegistrationDate='2023-13-01' (Month 13).
- Wrong Data Types (Structure):
 - Age='twenty'
- Inconsistent Formatting (Consistency):
 - Dates: Mixed 'YYYY-MM-DD', 'DD/MM/YYYY', 'Feb 3, 2023'.
 - Gender: Mixed 'F', 'Female', 'M', 'Male'.
 - Country: 'USA', 'US', 'United States'.
- Invalid Patterns (Content):
 - Emails like 'bob@invalid', 'hank@domain.'

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Missing data



- It is usual to define three kinds of missing data:
 - *Missing completely at random (MCAR)*
 - When missing values are randomly distributed across all observations, then we consider the data to be *missing completely at random*
 - *Missing at random (MAR)* – *most common type*
 - The key difference between MCAR and MAR is that under MAR the data is not missing randomly across all observations, but is missing randomly only within sub-samples of data.
 - *Missing not at random (MNAR)*
 - When the missing data has a structure to it, we cannot treat it as missing at random.

Missing data



MCAR		
ID	Gender	Depression Rating
1	Male	6
2	Male	2
3	Female	1
4	Male	4
5	Female	5
6	Female	9
7	Male	3
8	Female	4
9	Female	7
10	Male	8
Missing Value		

MAR		
ID	Gender	Depression Rating
1	Male	6
2	Male	2
3	Female	1
4	Male	4
5	Female	5
6	Female	9
7	Male	3
8	Female	4
9	Female	7
10	Male	8
Missing Value		

MNAR		
ID	Gender	Depression Rating
1	Male	6
2	Male	2
3	Female	1
4	Male	4
5	Female	5
6	Female	9
7	Male	3
8	Female	4
9	Female	7
10	Male	8
Missing Value		

How to deal with missing data

- There are two methods to solve the error:
 - **Deletion of data** – delete missing data to reduce bias
 - Dealing with data that is missing at random
 - **Listwise** – remove list of record
 - **Pairwise** – omit cases with MCAR data on the interested variables, but not for other analyses
 - **Entire variables** – omit an entire variable (column) from the analysis

Listwise Deletion				Pairwise Deletion - Color Sample				Entire Variable Deletion			
ID	Gender	Depression Rating	Favorite Color	ID	Gender	Depression Rating	Favorite Color	ID	Gender	Depression Rating	Favorite Color
1	Male	6	Blue	1	Male	6	Blue	1	Male	6	Blue
2	Male	2	Green	2	Male	2	Green	2	Male	2	NA
3	Female	1	Red	3	Female	1	Red	3	Female	1	Red
4	Male	4	NA	4	Male	4	NA	4	Male	4	NA
5	Female	5	Yellow	5	Female	5	Yellow	5	Female	5	NA
6	Female	9	Purple	6	Female	9	Purple	6	Female	9	NA
7	Male	3	Green	7	Male	3	Green	7	Male	3	Green
8	Female	4	Blue	8	Female	4	Blue	8	Female	4	NA
9	Female	NA	Blue	9	Female	NA	Blue	9	Female	7	NA
10	Male	8	Red	10	Male	8	Red	10	Male	8	NA
Deleted				Deleted				Deleted			

How to deal with missing data

- **Imputation of data** – guesses for missing data
 - Most useful when the percentage of missing data is low
 - **Mean/median/mode** – using the mean or median in place of the missing value
 - **Last Observation Carried Forward (LOCF)** – for *time-series data*, use the last observed value for any missing data
 - **Linear Interpolation** – for *time-series data*, use linear interpolation, where missing values are filled by considering the observations both before and after them

Mean Imputation		
ID	Gender	Depression Rating
1	Male	6
2	Male	2
3	Female	1
4	Male	4.75
5	Female	5
6	Female	9
7	Male	3
8	Female	4
9	Female	4.75
10	Male	8
Imputed		

LOCF Imputation		
Year	Country	Value
2016	DE	102
2017	DE	101
2018	DE	101
2019	DE	103
2020	DE	104
2016	FR	106
2017	FR	106
2018	FR	108
2019	FR	111
2020	FR	112
Imputed		

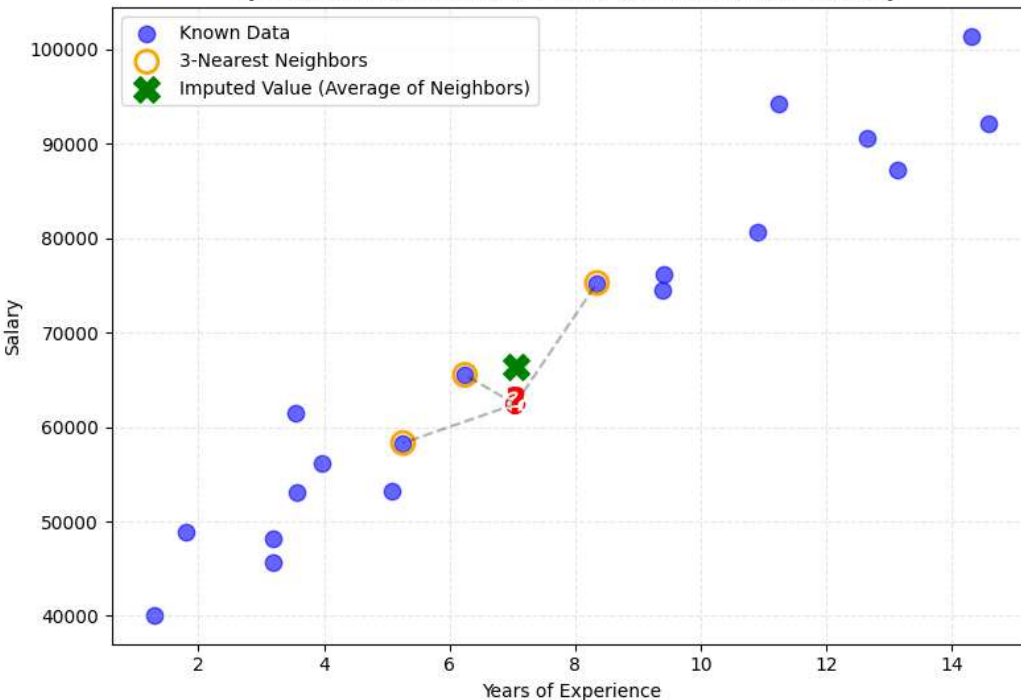
Linear Interpolation		
Year	Country	Value
2011	DE	95
2012	DE	94
2013	DE	96
2014	DE	98
2015	DE	100
2016	DE	102
2017	DE	101
2018	DE	102
2019	DE	103
2020	DE	104
Imputed		

How to deal with missing data

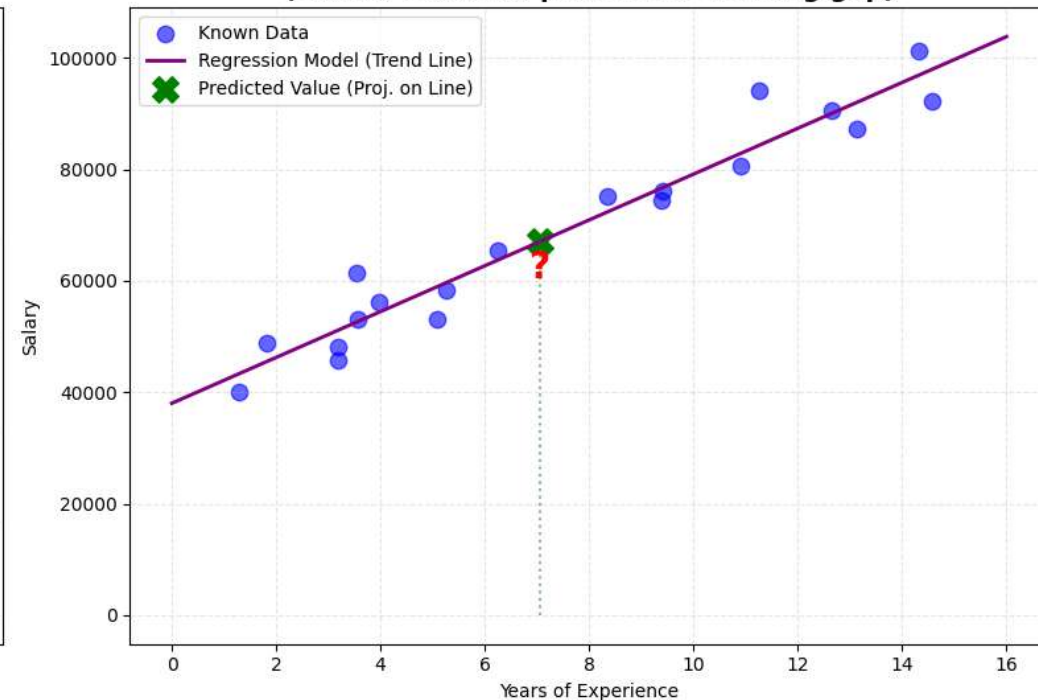
- **Advanced methods**

- **K-NN Imputation:** Find "similar" rows and borrow their values.
- **Prediction:** Train a model to predict the missing value.

Method 1: K-Nearest Neighbors (KNN)
(Find similar records and borrow their value)



Method 2: Machine Learning Prediction
(Train a model to predict the missing gap)



Average

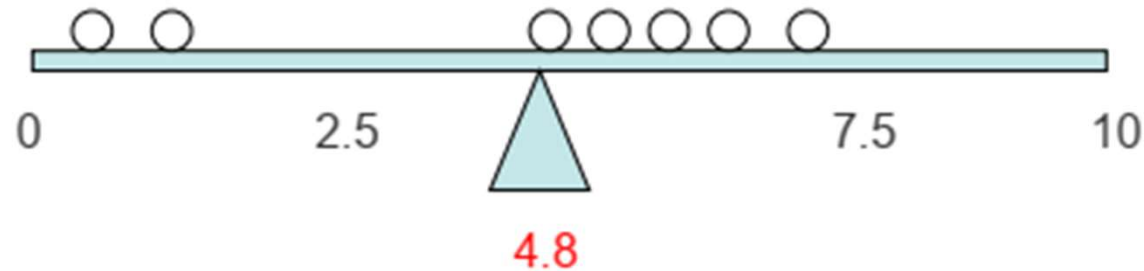
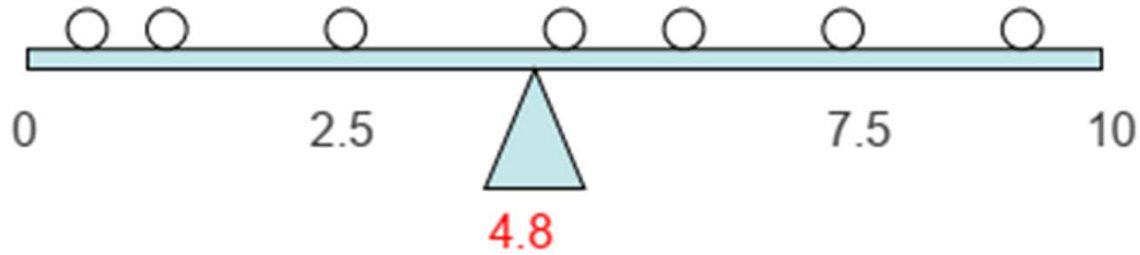
•Average (mean)

- Sum of all values divided by the number of values in the data set.
- One measure of a central location in the data set

73 – 66 – 69 – 67 – 49 – 60 – 81 – 71 – 78 – 62 – 53
– 87 – 74 – 65 – 74 – 50 – 85 – 45 – 63 – 100

$$\text{Average} = (73 + 66 + 69 + 67 + 49 + 60 + 81 + 71 + 78 + 62 + 53 + 87 + 74 + 65 + 74 + 50 + 85 + 45 + 63 + 100) = 68.6$$

The mean is vulnerable to problems



The data may or may not be symmetrical around its average value

Median



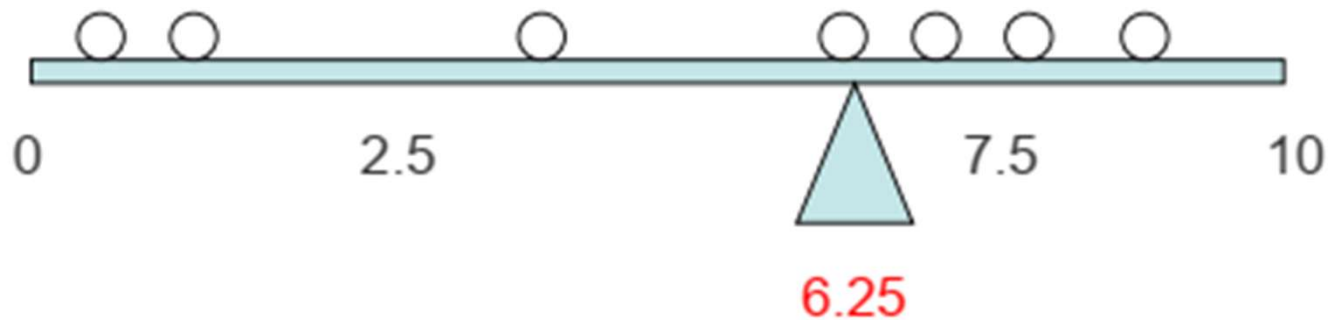
- The middle value in a sorted data set. Half the values are greater and half are less than the median.
- Another measure of a central location in the data set.

(45, 49, 50, 53, 60, 62, 63, 65, 66, 67, 69, 71, 73, 74, 74, 78, 81, 85, 87, 100)

Median: 68

Median

- May or may not be close to the mean.
- A combination of mean and median are used to define the skewness of a distribution.



Mode



- The most frequent occurring value.
- Another measure of central location in the data set.
- (45, 49, 50, 53, 60, 62, 63, 65, 66, 67, 69, 71, 73, 74, 74, 78, 81, 85, 87, 100)
- Mode: 74
- Generally not all that meaningful unless a larger percentage of the values are the same number





Q & A

